Multistage amplifiers

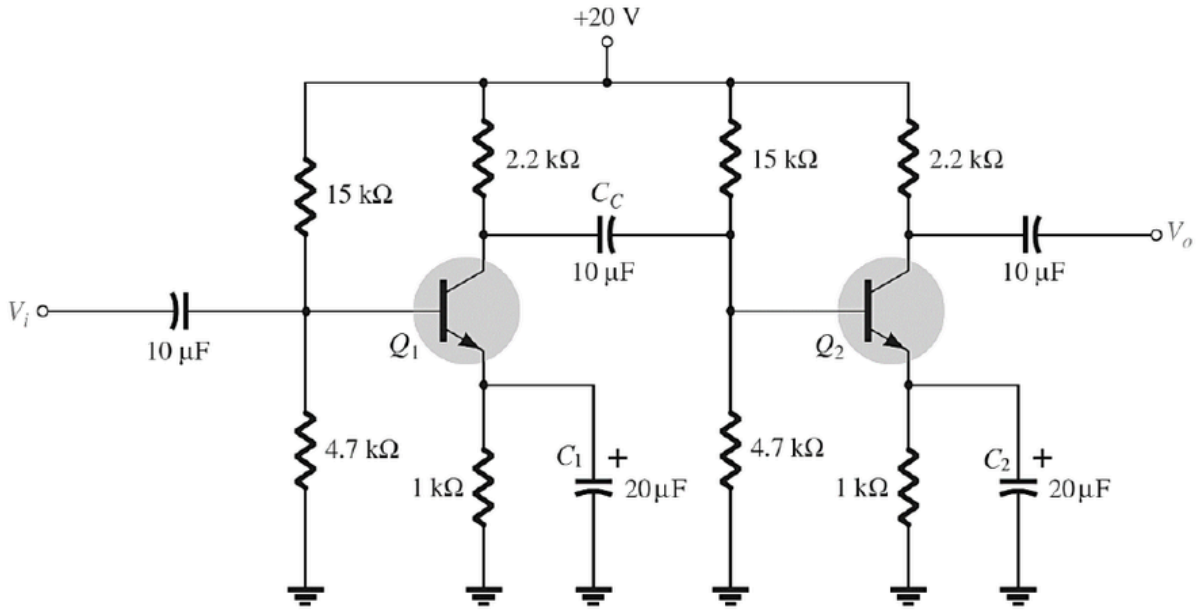
Practice questions · Day 14 · Fri 6 Nov · pack E8

5. Practice

From your course files

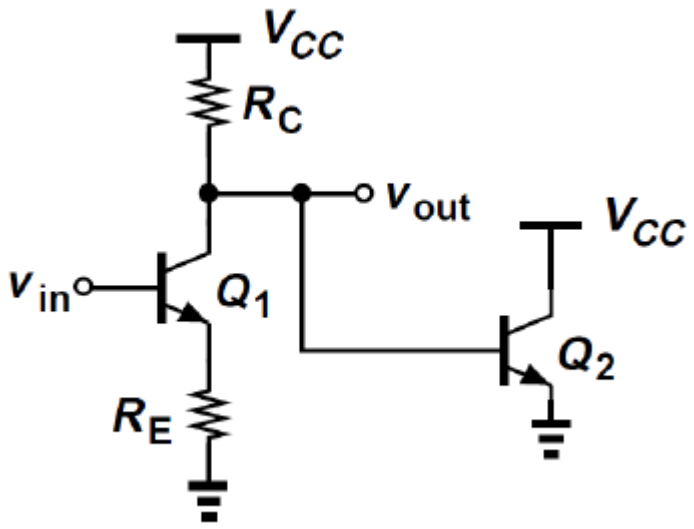
1. Prac 4 calculation section ("For the network of Figure 1: 1) Calculate the no-load voltage gain of the RC-coupled transistor amplifiers in Figure 1. 2) Calculate the overall gain and output voltage if a 4.7 kΩ load is applied to the second stage and compare it to the results of part (1). 3) Calculate the input impedance of the first stage and the output impedance of the second stage.") is written out in full as Worked Example 2 above. Redo it without looking: no-load 463, loaded 316, Z_{i_1} = 941 Ω, Z_{o_2} = 2.2 kΩ.

Prac 4 step 6 (practical). Add a bypass capacitor at the emitter of the second stage. The network should be the one in Figure 3. Calculate the voltage gain of the network of Figure 3.



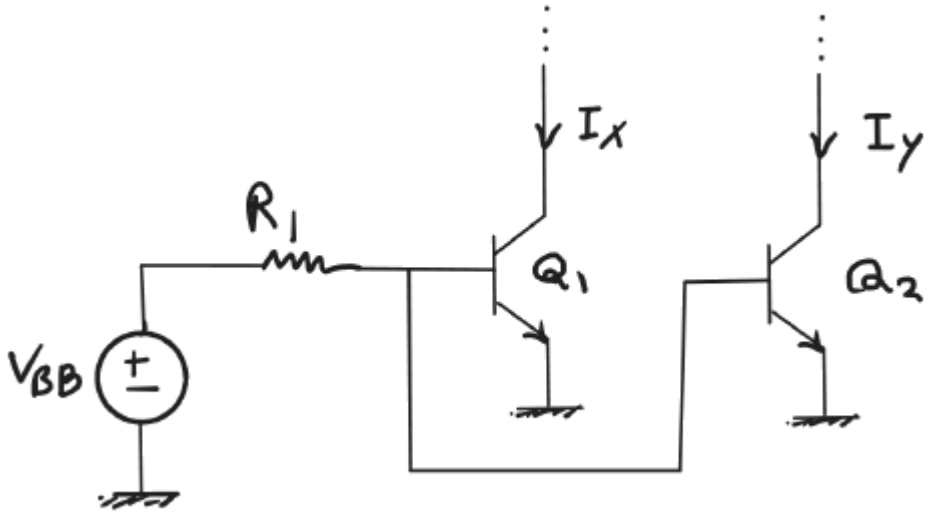
Prac 4 Figure 3 (from the practical handout)

WS6 Q10 (course question). The two BJTs in the amplifier of figure 10 are different. Using hybrid-π models, find an expression for AC voltage gain ($A_v = \frac{v_{out}}{v_{in}}$) of the amplifier.



WS6 Q10 figure (Figure 10 of the workshop sheet)

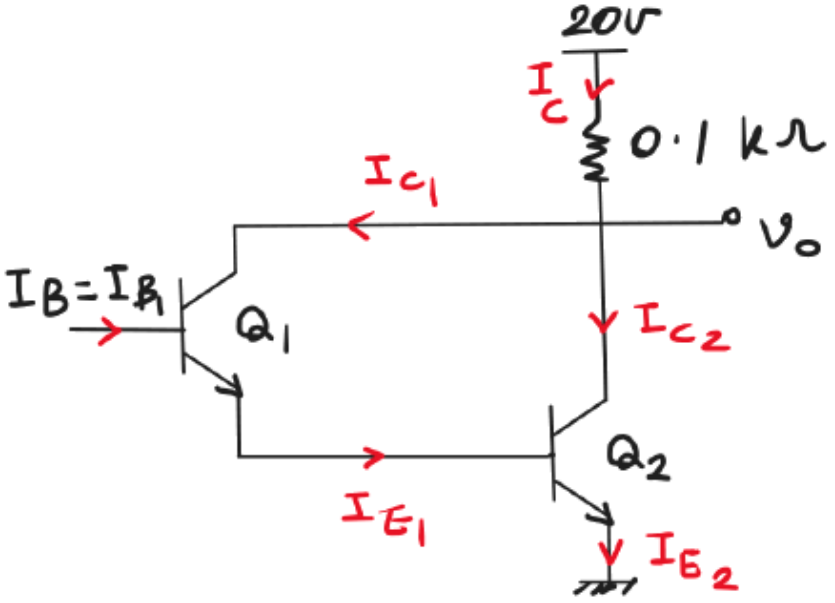
WS6 Q12 (course question). In the circuit shown in figure 12, the two BJTs are identical with $\beta = 100$, $V_{BE,on} = 0.7$ V and $V_{CE,sat} = 0.2$. If $I_X = 2I_Y = 1$ mA and $R_1 = 5\ k\Omega$, find voltage V_{BB}. Assume the two BJTs are in the active state.



WS6 Q12 figure (Figure 12 of the workshop sheet)

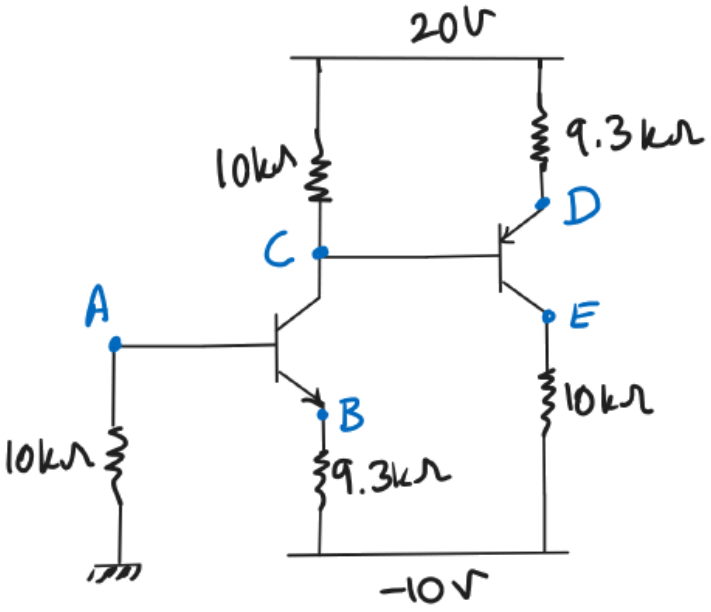
WS6 Q13 (course question). In the circuit shown in figure 13, $I_{E_2} = 120$ mA, $\alpha_1 = 0.99$, $\alpha_2 = 0.98$. Assuming that both transistors are in the **active** state, answer the following questions:

1. Find I_{C_1}, I_{B_1}, I_{E_1}, I_{C_2}, and I_C.
2. Find the output voltage, v_o.



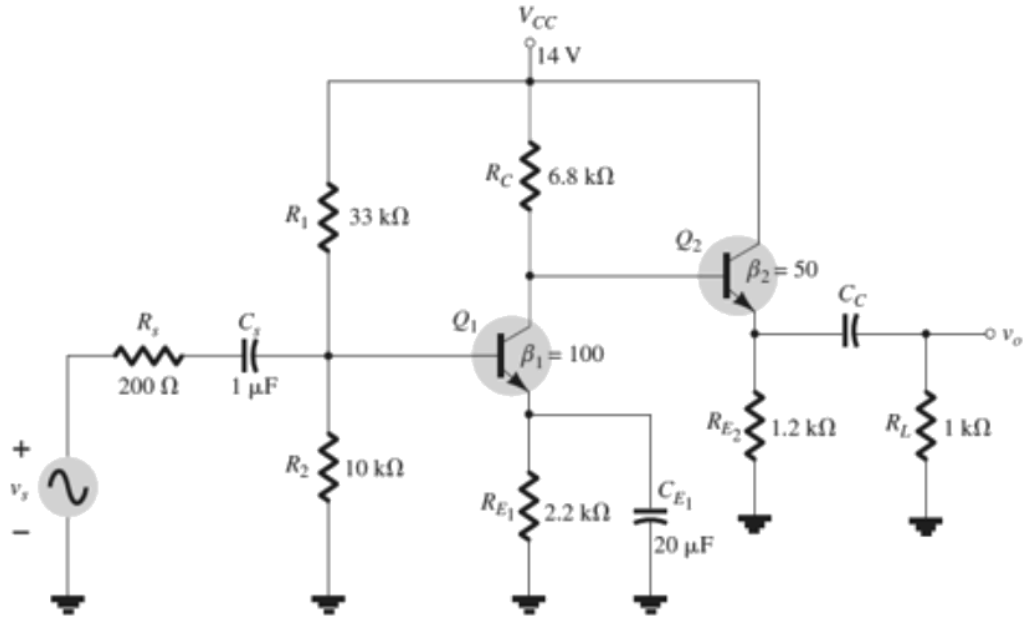
WS6 Q13 figure (Figure 13 of the workshop sheet)

WS6 Q14 (course question). In the circuit shown in Figure 14, both transistors have $V_{BE,on} = 0.7$ V. Assuming that $\beta = \infty$ and both transistors are in either **active or saturation** state, find the voltage at nodes A, B, C, D, and E.



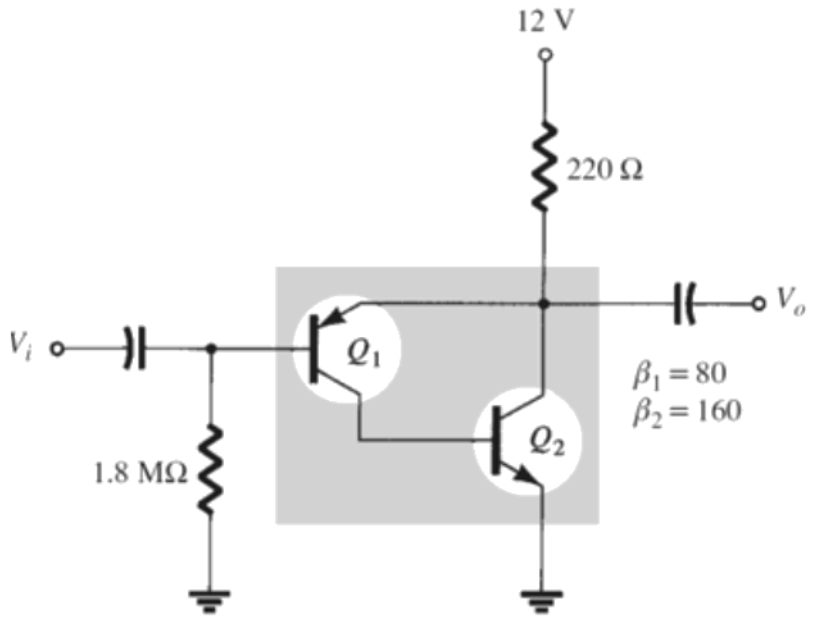
WS6 Q14 figure (Figure 14 of the workshop sheet)

WS6 Q16 (course question). Determine the dc levels for the currents and voltages of the direct-coupled amplifier of figure 16.



WS6 Q16 figure (Figure 16 of the workshop sheet)

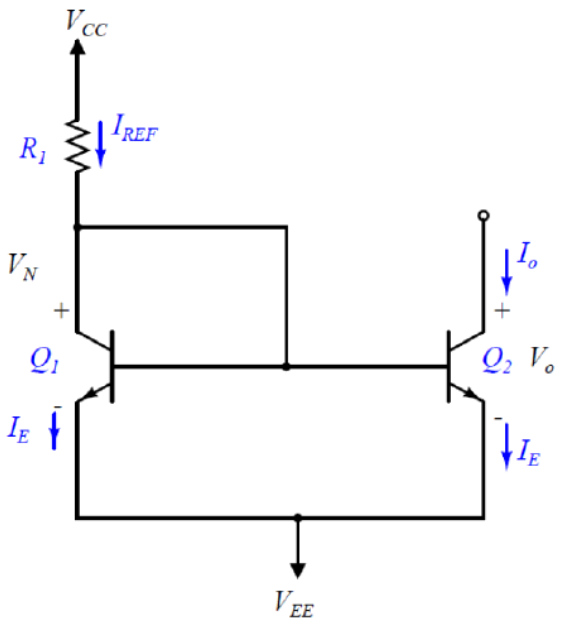
WS6 Q17 (course question). For the feedback amplifier of figure 17 determine (a) the base and collector current of each transistor and (b) the base, emitter, and collector voltages of each transistor.



WS6 Q17 figure (Figure 17 of the workshop sheet)

WS6 Q18 (course question). In the circuit below, a current mirror is shown where the transistors Q_1 and Q_2 are identical.

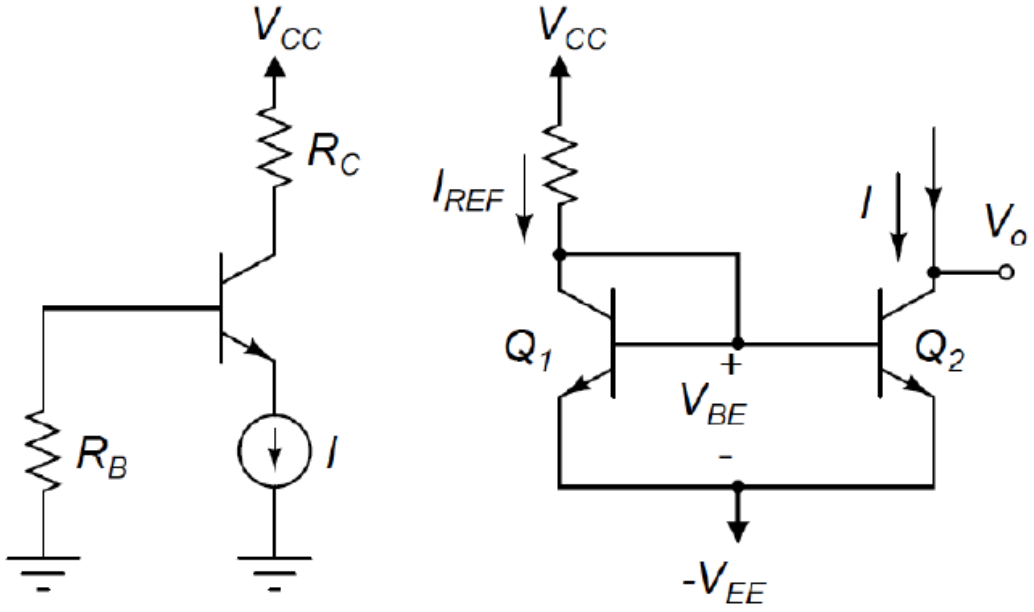
1. Assume the transistor β is not very large, derive an expression for the ratio of $\frac{I_o}{I_{REF}}$.
2. Simplify this expression if β is very large.



WS6 Q18 figure (from the workshop sheet)

WS6 Q19 (course question). In the circuit shown below, $V_{CC} = 10$ V, $I = 1$ mA, $\beta = 100$, $R_B = 100\ k\Omega$ and $R_C = 7.5\ k\Omega$

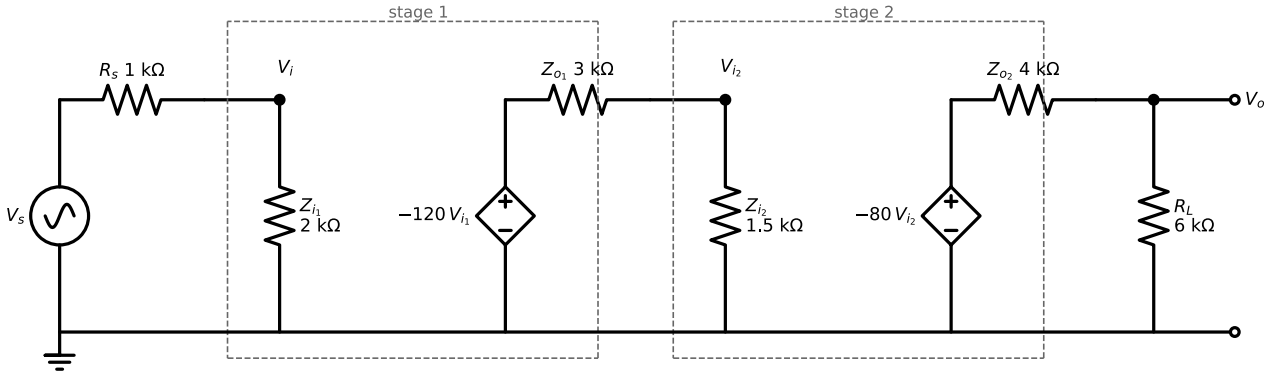
1. Find the DC voltage at the base, the emitter and the collector terminals of the left circuit.
2. Assuming $V_{EE} = 10$ V, and assuming β is very large, find the required value of the resistor in order for the right circuit to implement the current source I shown in the left circuit. Q_1 and Q_2 are identical.



WS6 Q19 figure (from the workshop sheet)

New questions

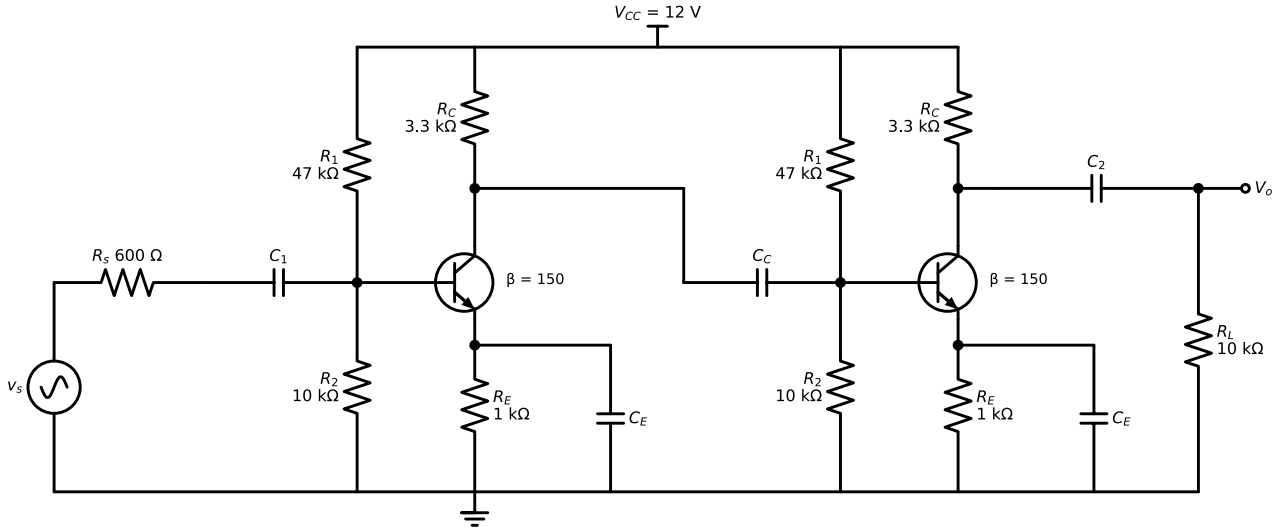
N1 [5 marks]. Two-port cascade. A two-stage amplifier is described by two-port parameters. Stage 1 has $A_{v_{NL1}} = -120$, $Z_{i_1} = 2$ kΩ, $Z_{o_1} = 3$ kΩ. Stage 2 has $A_{v_{NL2}} = -80$, $Z_{i_2} = 1.5$ kΩ, $Z_{o_2} = 4$ kΩ. The source has $R_s = 1$ kΩ and the load is $R_L = 6$ kΩ. (a) Find the loaded gain of each stage and $A_{v_T} = V_o/V_i$. [3] (b) Find $A_{v_s} = V_o/V_s$. [1] (c) By what factor is A_{v_s} below the "ideal" product $A_{v_{NL1}} A_{v_{NL2}}$? [1]



N1: the two-stage amplifier as a chain of two-port models

N2 [10 marks]. Two-stage CE–CE cascade. Two identical common-emitter stages are RC-coupled. Each stage uses voltage-divider bias from $V_{CC} = 12$ V, with $R_1 = 47$ kΩ from V_{CC} to the base and $R_2 = 10$ kΩ from the base to ground. Each has $R_C = 3.3$ kΩ from V_{CC} to the collector, and $R_E = 1$ kΩ from the emitter to ground, **fully bypassed** by a large capacitor. Both transistors have $\beta = 150$ and $r_o = \infty$. All coupling capacitors are shorts at signal frequency. A source with $R_s = 600\ \Omega$ drives the base of Q1. The collector of Q2 drives an $R_L = 10$ kΩ load through a coupling capacitor.

a. Find V_B, I_E, V_{CE} and r_e for each stage, justifying the approximate method. [3]
b. Find Z_{i_2} and the loaded gain of stage 1. [2]
c. Find the gain of stage 2 with R_L, and the overall $A_{v_T} = V_o/V_i$. [2]
d. Find Z_{i_1} and $A_{v_s} = V_o/V_s$. What is v_o for $v_s = 20$ µV peak? [3]



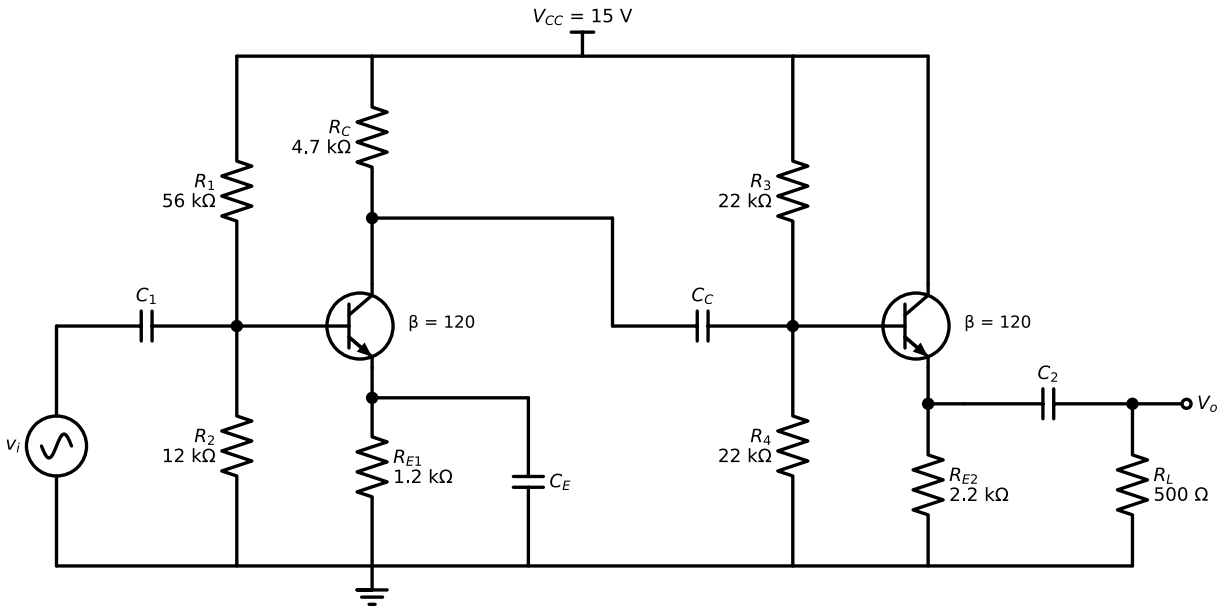
N2: two identical RC-coupled CE stages with R_s = 600 Ω and R_L = 10 kΩ

N3 [10 marks]. CE–EF cascade (buffering a low-resistance load). Stage 1 is a common-emitter amplifier on $V_{CC} = 15$ V with $R_1 = 56$ kΩ (to V_{CC}), $R_2 = 12$ kΩ (to ground), $R_C = 4.7$ kΩ, and $R_{E1} = 1.2$ kΩ fully bypassed. Its collector is capacitor-coupled to the base of stage 2. Stage 2 is an emitter follower on the same 15 V supply, with base divider $R_3 = 22$ kΩ (to V_{CC}) and $R_4 = 22$ kΩ (to ground), collector tied directly to V_{CC}, and $R_{E2} = 2.2$ kΩ from emitter to ground. The emitter of Q2 drives an $R_L = 500\ \Omega$ load through a coupling capacitor. Both transistors have $\beta = 120$ and $r_o = \infty$.

a. Find I_{E1}, r_{e1}, I_{E2} and r_{e2}. [3]
b. Find Z_{i_2}. [2]

c. Find A_{v_1}, A_{v_2} and A_{v_T}. [3]

d. Find the gain if the 500 Ω load were connected straight to Q1's collector (no follower), and explain the difference. [2]



N3: CE stage capacitor-coupled to an emitter follower that drives the 500 Ω load

N4 [7 marks]. Darlington emitter follower. A Darlington pair (Q1 emitter to Q2 base, collectors tied together to V_{CC}) is biased from $V_{CC} = 18$ V by $R_B = 3.3$ MΩ from V_{CC} to Q1's base. An emitter resistor $R_E = 390\ \Omega$ runs from Q2's emitter to ground. $\beta_1 = 100$, $\beta_2 = 80$, and the Darlington $V_{BE} = 1.6$ V. (a) Find β_D, I_B, I_E, V_E and V_{CE}. [4] (b) Find Z_i and A_i (no load), and state A_v. [3]

N4: Darlington emitter follower

N5 [6 marks]. Current mirror design. Two identical npn transistors form a basic current mirror. Both emitters are grounded and the bases are tied together. Q1 is diode-connected, and R_{REF} runs from $V_{CC} = 12$ V to Q1's collector. Q2's collector draws the output current I_o through a load resistor R_L from the same 12 V rail. $V_{BE} = 0.7$ V, $V_{CE,\text{sat}} = 0.2$ V and $\beta = 80$.

a. Choose R_{REF} for $I_o = 2$ mA, assuming β is very large. [1]
b. With that resistor and $\beta = 80$, what is the actual I_o? [2]
c. What R_{REF} gives exactly 2 mA with $\beta = 80$? [2]
d. What is the largest R_L that keeps Q2 active? [1]

N5: basic current mirror with R_{REF} to be designed and load R_L on Q_2

6. Quick recall check

1. In a two-stage cascade, which stage's gain changes when you connect stage 2 to stage 1, and what resistance causes it?
2. Write Z_{i_2} for a stage 2 that is a CE stage with an **unbypassed** R_E.
3. A Darlington has $\beta_1 = 50$ and $\beta_2 = 120$. What are β_D and the approximate Z_i (ignoring R_B) with $R_E = 1$ kΩ?
4. Why does a cascode have a wider bandwidth than a single CE stage with the same gain?
5. A current mirror has $\beta = 50$. What is I_o/I_{REF}?